Ex. No. 2 **LOGICAL OPERATIONS USING 8086**

AIM

To write an assembly language program to implement

(a) AND GATE
(b) OR GATE
(c) Half Adder

SOFTWARE

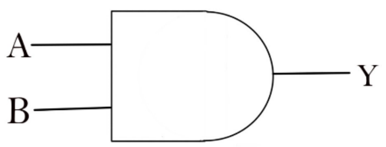
8086 Microprocessor online simulator software.

PROCEDURE

1. Start 8086 online simulator software and enter the program.
2. Compile the program and check for any errors in the program.
3. Enter the input data’s, compile and run the program.
4. Check the results in the specified registers.

(a) AND GATE

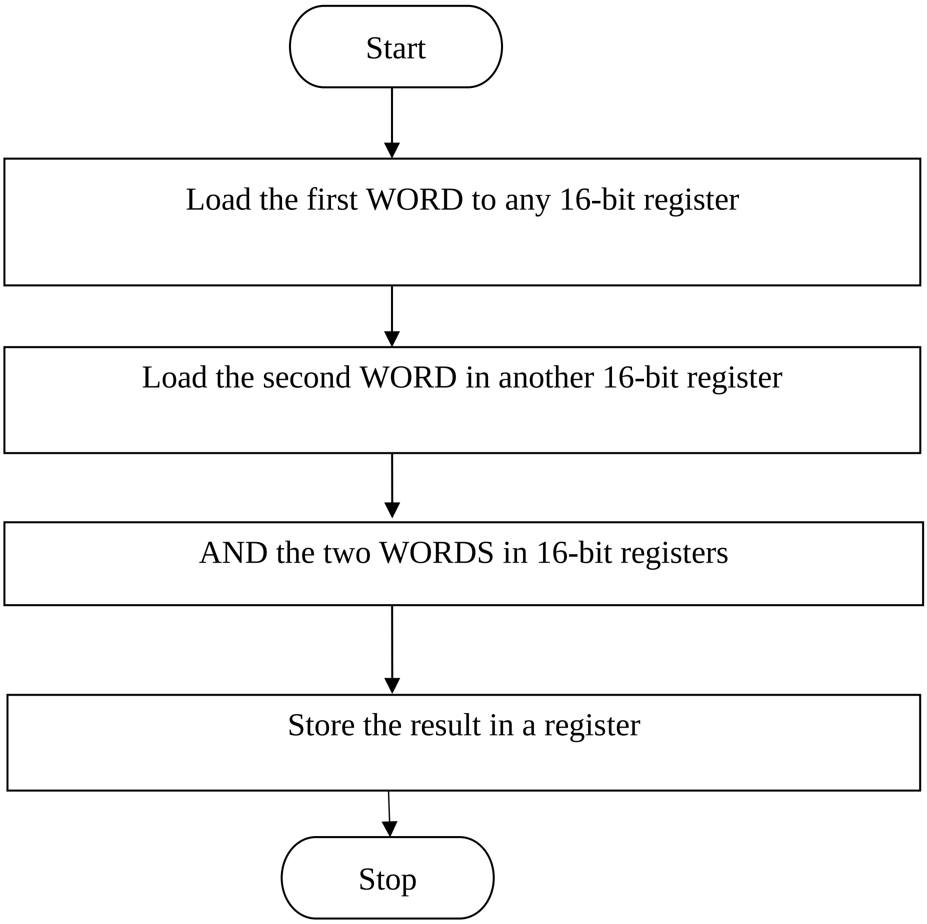
SYMBOL



TRUTH TABLE

INPUT		OUTPUT
A	**B**	**Y**
0	0	0
0	1	0
1	0	0
1	1	1

FLOW CHART



TABULATION

Input Data

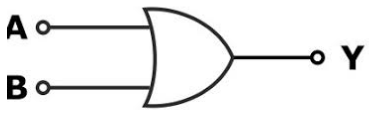
Registers		
Data1		
Registers		
Data 2		

Output data

Register		
Data		

(b) OR GATE

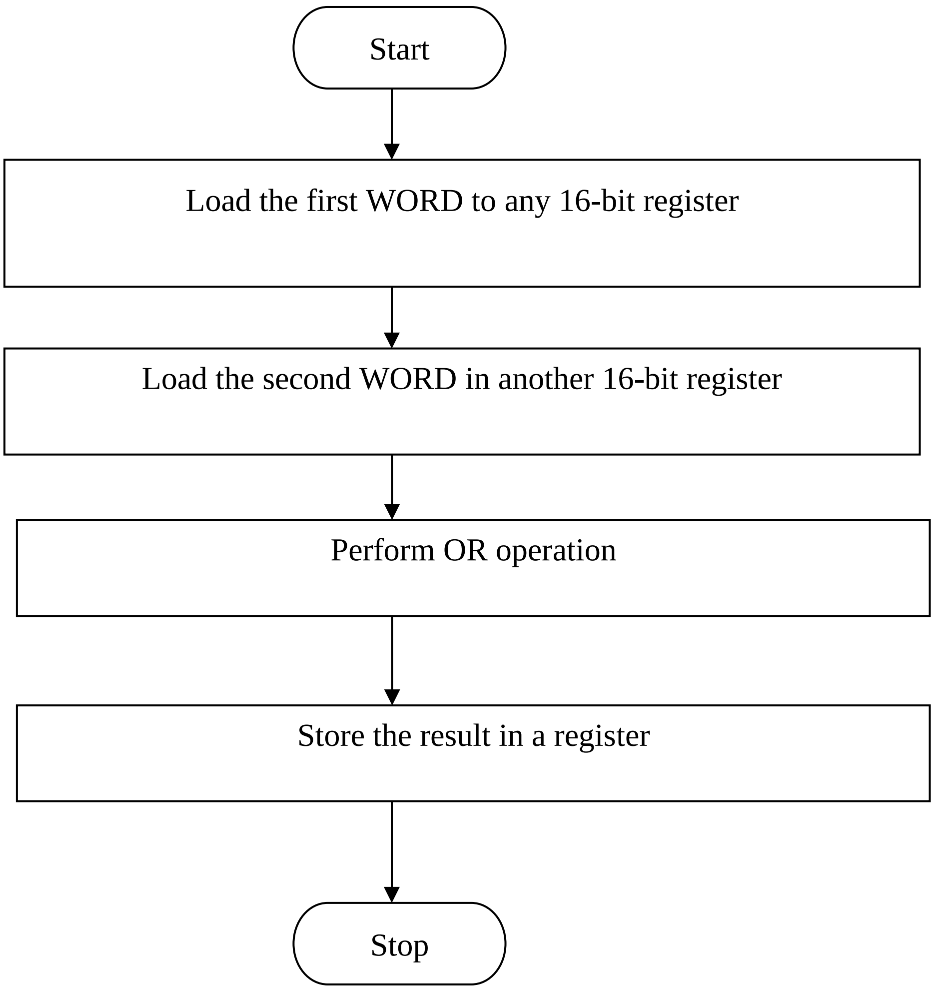
SYMBOL



TRUTH TABLE

INPUT		OUTPUT
A	**B**	**Y**
0	0	0
0	1	1
1	0	1
1	1	1

FLOW CHART



TABULATION

Input data

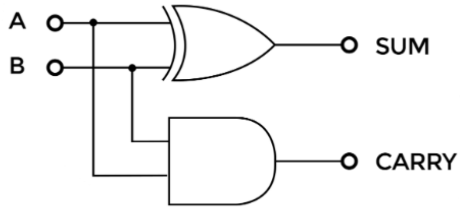
Registers		
Data1		
Registers		
Data 2		

Output data

Register		
Output Data		

(c) HALF ADDER

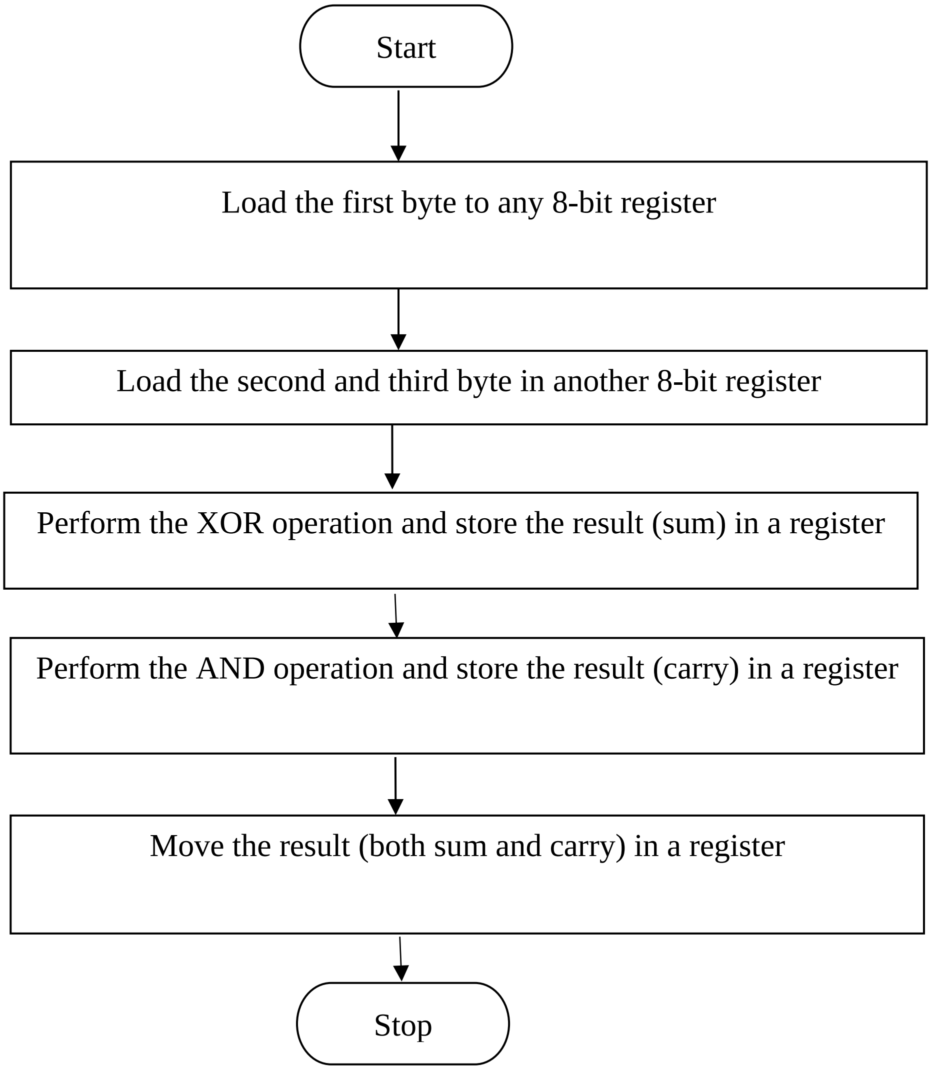
CIRCUIT DIAGRAM



TRUTH TABLE

INPUT		OUTPUT	
A	**B**	**SUM**	**CARRY**
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

FLOWCHART



TABULATION

Input data

Registers		
Data1		
Registers		
Data 2		

Output data

Register		
Output Data (Sum)		
Output Data (Carry)		

RESULT

Thus, the logical operations were performed using 8086 Microprocessor.